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Detectability of Graviational Waves from Primordial Black Hole Mergers with the ALPS II Experiment

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Abstract

In this study, the possible signal that a Primordial Black Hole (PBH) binary merger could produce at the Any Light Particle Search II (ALPS II) is analysed. Primordial Black Holes are predicted to have formed in the early universe and are still a promising dark matter candidate in certain mass ranges. The ALPS II is a light-shining through a wall (LSW) experiment that searches for axions or axion like particles. A strong magnetic field and highly sensitive photon detection scheme open up ALPS II to the possibility of detecting ultra-high frequency gravitational waves. To this end the gravitational wave signal of a PBH merger is computed. Then the conversion of the gravitational waves to electromagnetic waves in an external magnetic field through graviton-photon mixing is calculated. For the detection of such a signal, highly sensitive photon detectors are required. The Superconducting Nanowire Single Photon Detector (SNSPD) and the Transition Edge Sensor (TES) are superconducting detectors that could fill this role. The observed photon signal and the detection significance from the merger at ALPS II is calculated for the case of the TES. It is determined that the gravitational wave signal of a possible PBH merger is too weak to allow for a sufficient detection significance at ALPS II.

Resumé

I denne undersøgelse analyseres det mulige signal, som en primordial sort hul (PBH) binær fusion kunne producere ved Any Light Particle Search II (ALPS II). Primordiale sorte huller forventes at være dannet i det tidlige univers og er stadig en lovende kandidat til mørkt stof i visse masseområder. ALPS II er et eksperiment, hvor lyset skinner gennem en væg (LSW), og som søger efter axioner eller axionlignende partikler. Et stærkt magnetisk felt og et meget følsomt fotondetektionsskema åbner op for muligheden for at detektere ultrahøjfrekvente gravitationsbølger. Til dette formål beregnes gravitationsbølgesignalet fra en PBH-fusion. Derefter beregnes omdannelsen af gravitationsbølgerne til elektromagnetiske bølger i et eksternt magnetfelt gennem graviton-foton-blanding . For at opdage et sådant signal kræves der meget følsomme fotondetektorer . Superconducting Nanowire Single Photon Detector (SNSPD) og Transition Edge Sensor (TES) er superledende detektorer, der kan udfylde denne rolle. Det observerede fotonsignal og betydningen af detektionen fra fusionen ved ALPS II er beregnet for TES'en. Det fastslås, at gravitationsbølgesignalet fra en mulig PBH-fusion er for svagt til at give mulighed for en tilstrækkelig detektionsværdi på ALPS II.

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1 Introduction

Of the entirety of matter inside the universe only a small fraction is baryonic. As several cosmological and astronomical observations show about 85 % [13] is in the form of cold dark matter (DM). Over time numerous explanations have formed to explain what this DM could possibly be. Of those, many assume the existence of a new type of particle whose interactions with the standard model have not been observed as of yet. These particles are of particular interest as they open up options for new kinds of physics beyond the standard model.

A different kind of candidate for DM however finds itself in the form of primordial black holes (PBHs). These black holes are assumed to have formed as a result of density fluctuations in the early universe. Some over dense regions collapse under the gravitational forces and create a PBH. The masses of the PBHs depend on the size of the region at the time of collapse. This allows for a possible wide spectrum of PBH masses at the time of their creation. While the lightest black holes would have evaporated over the lifetime of the universe through Hawking radiation those with masses above 10^{15} g would still be around today. Of those remaining many masses are already ruled out, but especially in the mass range of $10^{-18} - 10^{-11} M_{\odot}$ they are a prime candidate to make up a substantial fraction of DM.

One option to search for PBHs is to look for possible gravitational wave signals. As they are a lot lighter than black holes formed after the collapse of heavy stars their gravitational waves are of much higher frequencies. This requires different kinds of detection methods compared to current large interferometers like LIGO, VIRGO, or KARGA. One possible idea is to use so called "Light Shining through Wall" experiments for this purpose. The ALPS II experiment located at DESY ("Deutsches Elektronen-Syonchrotron") in Hamburg is one such experiment. It is trying to detect another prime candidate for DM, so called axions or axion like particles (ALPs). These are very weakly interacting particles and can possibly convert into light in the presence of strong magnetic fields [23].

Inside the strong magnetic field of ALPS II gravitational waves might also convert to photons through graviton to photon mixing [11]. This way an experiment used for the detection of ALPs could be used to simultaneously search for a different kind of DM in PBHs. In this study the question will be answered whether the ALPS II experiment could detect a PBH merger. So first the gravitational wave signal of a merger at certain distance from earth is calculated. Afterwards the conversion of the gravitational wave to a photon signal inside the magnetic field of ALPS II is analysed. In order to detect such a signal highly sensitive and low noise detectors are required.

Two such sensors, the Transition Edge Sensor (TES) and the Superconducting Nanowire Single-photon Detector (SNSPD) will be looked at. As the name implies the SNSPD relies on a superconductive wire which requires operational temperatures of below 10 K to function. To this end it must be operated within a cryostat using circulating Helium to reach a consistent base temperature of around 1.5 K. The TES will make a substantial point of discussion in this study since its efficiency dictates the limit for PBH detection.

2 Primordial Black Holes

Black holes that are created through astrophysical processes, such as the collapse of a dying star or a merger between two heavy objects, have a minimum mass of about three solar masses [3]. While these objects are well described theoretically and observed experimentally, there is a potential different type of black hole that poses more of a challenge. These so-called primordial black holes (PBH) are hypothesised to have their origin in the early universe and are, because of their wide range of possible masses as well as the characteristics they share with other black holes, popular dark matter candidates [13]. The possibility of their existence has been discussed as early as the 1970s, but their primordial origin and large potential mass spectrum has made the direct observation difficult and the theoretical description uncertain. This Chapter will be a brief introduction of this extensive topic and discuss key features of PBH creation in Section 2.1 and their mass spectrum in Section 2.2. Lastly the binary merger of two PBHs and the resulting gravitational wave signal will be discussed (Section 2.3).

2.1 Formation of PBHs

An in depth explanation of PBH formation can be found in Ref. [26], of which a short resume is given here. The formation of PBHs would have taken place only a short time after the Big Bang, when the universe was still radiation-dominated. Generally, assuming a Friedmann metric and therefore space to be flat and homogeneous, the expansion would have counteracted the gravitational energy of any given region. But if, for any of the numerous possible reasons, local density fluctuations happened, so-called overdense regions could have formed. Because of their high density, they would have collapsed when their potential became greater than the kinetic expansion energy, forming PBHs. It is likely that such regions existed, because similar fluctuations are assumed to explain for example the formation of galaxies. The PBHs that were created in this way could have had almost no lower bound to their potential mass, with the lightest possible being at less than a Planck mass, so of order 10^{-8} kg.

2.2 Mass Spectrum

There are multiple constraints on the masses of possible PBHs, both theoretical constraints in the lower limit and experimental observations ruling out specific masses. It is necessary to differentiate between the constraints on the initial masses of PBHs and the masses that are possible to observe today. The first point is certainly important and influences many processes in the early universe that still influence astrophysical phenomena. Nevertheless, the focus of this discussion lies on the second point because of the relevance for the possible observation of PBHs today. This section follows closely Refs. [7] and [30]. Black holes are theorized to evaporate by radiating particles like a

black body of temperature

$$T \approx 10^{26} \left(\frac{M_{PBH}}{\mathrm{g}} \right)^{-1} \mathrm{K}, \tag{2.2.1}$$

assuming that the black hole has neither charge nor angular momentum. Through this so-called Hawking radiation, black holes lose more and more mass while their temperature rises. Assuming that PBHs were created shortly after the Big Bang, only the ones with an initial mass of at least

$$M_{\mathrm{crit}} = 10^{15} \mathrm{g} \tag{2.2.2}$$

would have survived to this day [26]. This is therefore the lower mass limit for observable PHBs. The black holes which at formation were slightly heavier than the critical mass would fully evaporate today, which could be measurable as bursts of radiated particles. There is some debate on whether black holes can evaporate completely or if there are stable remnants left after most of the mass is lost through radiation (see Ref. [4]), but this discussion is, though certainly interesting, beyond the scope of this thesis.

In Ref.[6] a detailed overview of the important constraints on the fraction of dark matter in the form of PBHs is given. The combination of the compiled results are shown in Figure2.1 taken from [6]. In the category of non-evaporated PHBs, it is most useful to express the constraints on the masses as the fraction of dark matter that would be composed of PHBs $f(M)$. Currently there are only a few mass ranges where PBHs could explain all the dark matter. In many of the possible mass ranges experiments were performed that either completely exclude them because no PBHs were observed or make them at least highly unlikely.

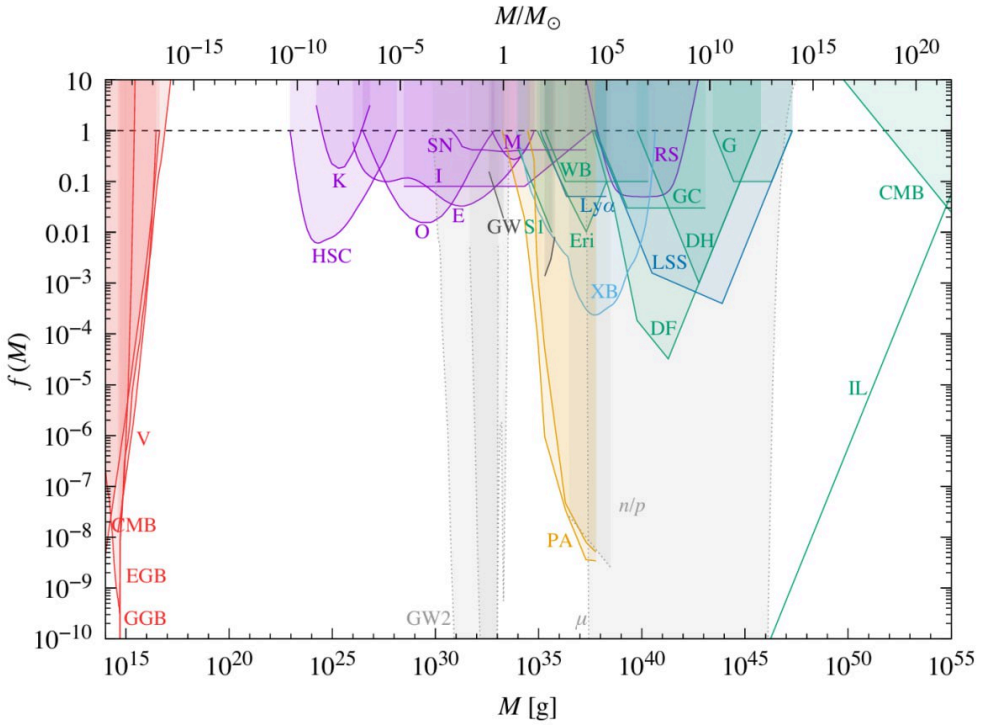


Figure 2.1: Experimental constraints on the fraction of dark matter in the form of PBHs $f(M)$ plotted for the PBH mass in g and in solar masses (from [6]).

In Fig. 2.1, the red areas correspond to the already discussed constraint due to evaporation. The following are the higher mass constraints:

- The purple areas are ruled out by observing no appropriate lensing effects, for example from measurements using stars from the Magellanic Clouds or fast radio bursts.
- The green areas correspond to dynamical constraints, more explicitly the potential destruction by collision, dynamical friction, passing, or consumption of different astronomical objects, for example wide binary star systems, neutron stars and white dwarfs, by passing PBHs. As the survival rate of these objects is known and the potential rate of destruction can be calculated from the known dark matter density in different regions of space, this poses a constraint on $f(M)$.
- The presence of PBHs would also effect the existence, formation, and structure of large scale structures such as galaxies and clusters of galaxies. The observation of the actual objects, again in combination with the knowledge of the dark matter distribution, poses another set of constraints on $f(M)$, displayed in dark blue.
- The light blue areas correspond to constraints due to the accretion of PBHs because of the limited possible growth during the early, radiation-dominated phase of the universe. The universe today also poses constrains on the maximum possible size of PBHs, but because this heavily depends on the location of the black hole and its surrounding, this is not shown in Fig. 2.1.
- The Hawking radiation of evaporating PBHs in the early universe should have had an effect on the cosmic microwave background, the ionisation and the temperature development of the universe due to the emitted particles. The constraints imposed by this are shown in orange.
- The black curve corresponds to the constraint due to the merger rates observed by LIGO.
- The gray areas are the constraints due to gravitational wave background, caused directly by the presence of PBHs, and μ distortions, special distortions of the CMB due to a specific shift in the state of the early universe.

It is important to note that the mass range the ALPS II experiment is sensitive to, $M_{PBH} < 10^{-11} M_{\odot}$ is not excluded by any of the constraints.

2.3 PBH merger

While many masses of primordial black holes are already ruled out there is still the possibility to detect them in certain mass ranges. This poses the question on how one could possibly search for these hypothetical black holes. One option for detection is Hawking radiation emitted during the process of evaporation. This is particularly interesting for light PBHs as those have a higher black body temperature and emit more radiation. Equally as interesting however is the possibility of PBHs emitting gravitational

waves. The first direct detection of gravitational waves from a black hole binary merger back in 2015 by the LIGO observatories [1] showed the potential for gravitational waves as sources to observe black holes. As the possible masses for the PBHs can be much lighter than black holes formed after the collapse of stars, the emitted gravitational waves are of very high frequency. This opens up the possibility for new kinds of detection making use of photon to graviton conversion inside large magnetic fields as will be discussed in Chapter 3.

This study will focus on the possible mergers of PBHs. Therefore, it is important to analyse in more detail the gravitational wave signal of such a merger. The formulas used here are taken from [18] by M. Maggiore. We consider the case of two orbiting PBHs of masses m_1 and m_2, in the Newtonian approximation. Additionally, the black holes can be regarded as point like which is a reasonable assumption as long as they are not yet too close to each other. For simplicity, first it is assumed that the black holes orbit each other in a stable circular orbit. By using the centre of mass frame, the problem is reducible to a one body problem for the reduced mass $\mu = \frac{m_1 m_2}{m_1+m_2}$. The orbital frequency ω_s can then simply be calculated and depends on the cube of the orbital radius R:

$$\omega_s^2 = \frac{G(m_1+m_2)}{R^3}. \tag{2.3.1}$$

Following [18] for two masses orbiting each other with an unchanging radius the strain for the two polarisations of the gravitational wave can be calculated. At a time t for an observer at distance r, they are given as

$$\begin{aligned} h_+(t) &= \frac{4}{r}\left(\frac{GM_c}{c^2}\right)^{5/3}\left(\frac{\pi f_{gw}}{c}\right)^{2/3}\frac{1+\cos^2\theta}{2}\cos(2\pi f_{gw}t), \\ h_\times(t) &= \frac{4}{r}\left(\frac{GM_c}{c^2}\right)^{5/3}\left(\frac{\pi f_{gw}}{c}\right)^{2/3}\cos\theta\sin(2\pi f_{gw}t). \end{aligned} \tag{2.3.2}$$

Here M_c is the so called chirp mass and is defined as

$$M_c = \mu^{3/5}(m_1+m_2)^{2/5} = \frac{(m_1m_2)^{3/5}}{(m_1+m_2)^{1/5}}. \tag{2.3.3}$$

Note that the frequency of the gravitational wave f_{gw} is related simply to the orbital frequency ω_s through

$$2\pi f_{gw} = \omega_{gw} = 2\omega_s. \tag{2.3.4}$$

The angle θ is the binary orientation angle between the rotation axis of the binary and the direction from the source to the observer and is shown in Figure 2.2.

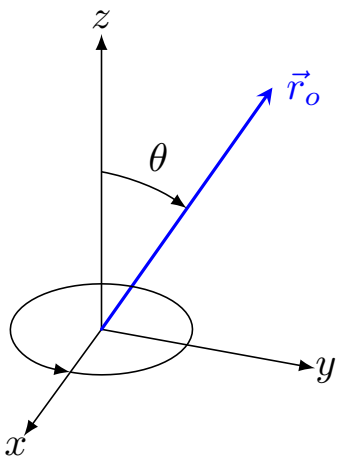


Figure 2.2: The coordinate system is chosen so that the rotating black holes find themselves in the x, y-plane. The binary angle θ is the opening angle between the rotation axis (in z-direction) and the direction to an observer $\vec{r}_o$.

In the current picture it is assumed that the radius between the orbiting black holes is constant. However since gravitational waves are emitted, the system loses energy at a rate equal to the radiated power. The total energy loss through radiation can be calculated by first determining the power radiated in any direction and then integrating over the solid angle. The total radiated power (see [18]) turns out to be

$$P = \frac{32}{5}\frac{c^5}{G}\left(\frac{GM_c\omega_{gw}}{2c^3}\right)^{10/3}. \tag{2.3.5}$$

The orbital energy E_{orbit} of the system depends on the kinetic energy of the black holes and the gravitational potential. Making use of the Virial theorem E_{orbit} is equal to half the potential energy. Using Eq. (2.3.1) and the chirp mass from Eq. (2.3.3) it can be expressed through the orbital frequency as follows:

$$E_{orbit} = E_{kin} + E_{pot} = -\frac{Gm_1m_2}{2R} = -\frac{G}{2}^{2/3} M_c^{5/3}\omega_s^{2/3}. \tag{2.3.6}$$

For a loss in energy, R needs to decrease which directly causes an increase in ω_s. If one assumes an inspiral phase with $\dot{\omega}_s \ll \omega_s^2$ then the motion can be approximated as quasi-circular (circular orbit with slowly decreasing radius). Thus the energy change $\frac{dE_{orbit}}{dt} = -P$ gives a relation between ω_s and $\dot{\omega}_s$ (and consequently for f_{gw} and $\dot{f}_{gw}$). Integrating the resulting differential equation over time gives an expression for $f_{gw}(t)$:

$$f_{gw}(\tau) = \frac{1}{\pi}\left(\frac{5}{256}\frac{1}{\tau}\right)^{3/8}\left(\frac{GM_c}{c^3}\right)^{-5/8}, \tag{2.3.7}$$

with τ the time to coalescence ($\tau = t_{coal} - t$). In Figure 2.3, $f(\tau)$ is plotted for two equal mass PBH mergers with different total mass. The heavier the system the faster the change in frequency. This is important for observation, since one is always limited to a frequency band for detecting a signal. Thus, while the strain of the gravitational waves is weaker, the signal duration is longer for lighter mass PBHs.

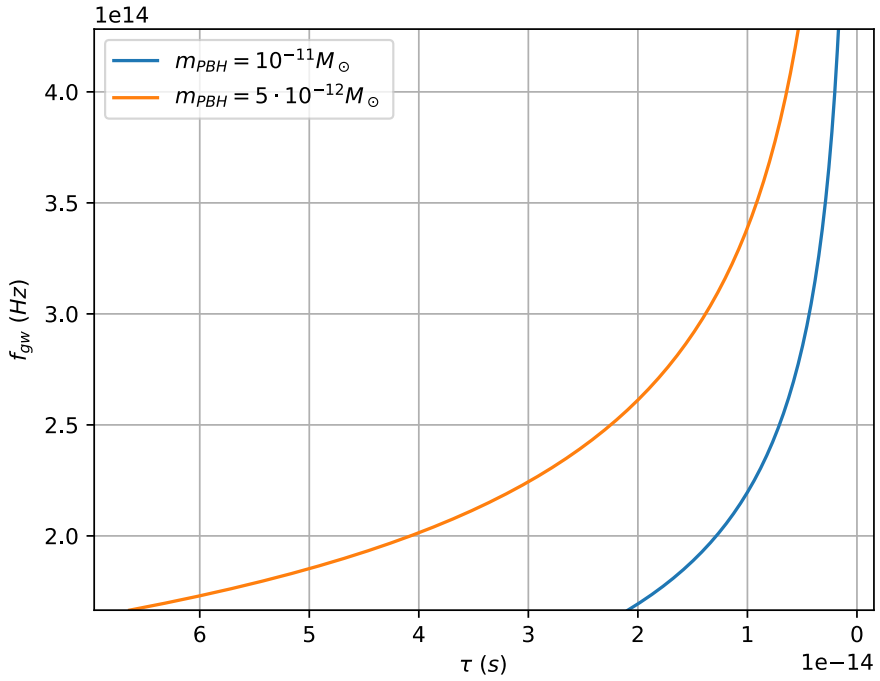


Figure 2.3: The gravitational wave frequency for two different equal mass PBH binaries as a function of time until coalescence.

The frequency diverges when τ goes to zero. With the increasing frequency the radius decreases. When the two masses get so close to each other that the actual merger takes place, the Newtonian approximation does not hold any more. The radius for which stable circular orbits are no longer possible is given as

$$r_{ISCO} = \frac{6G(m_1 + m_2)}{c^2}. \tag{2.3.8}$$

The corresponding orbit is called the Innermost Stable Circular Orbit (ISCO). From r_{ISCO} one can calculate a maximum orbital frequency following Eq. (2.3.1) for which the Newtonian approximation still holds. The corresponding maximum possible gravitational wave frequency f_{max} is given as

$$f_{max} = \frac{1}{6\sqrt{6}\pi} \frac{c^3}{G(m_1 + m_2)}. \tag{2.3.9}$$

This frequency is especially interesting for observation as it limits the mass ranges for which a detection in the inspiral phase is still possible. As will be discussed in more detail in Chapter 4, the ALPS II experiment is only sensitive to frequencies in the order of 100 THz. If the masses of a binary is too large, f_{max} will be too small and thus it can not be detected with the ALPS II experiment. For equal mass binaries the maximum frequency is shown in Figure 2.4. Also indicated is the upper limit for masses that can not be detected by the ALPS II experiment.

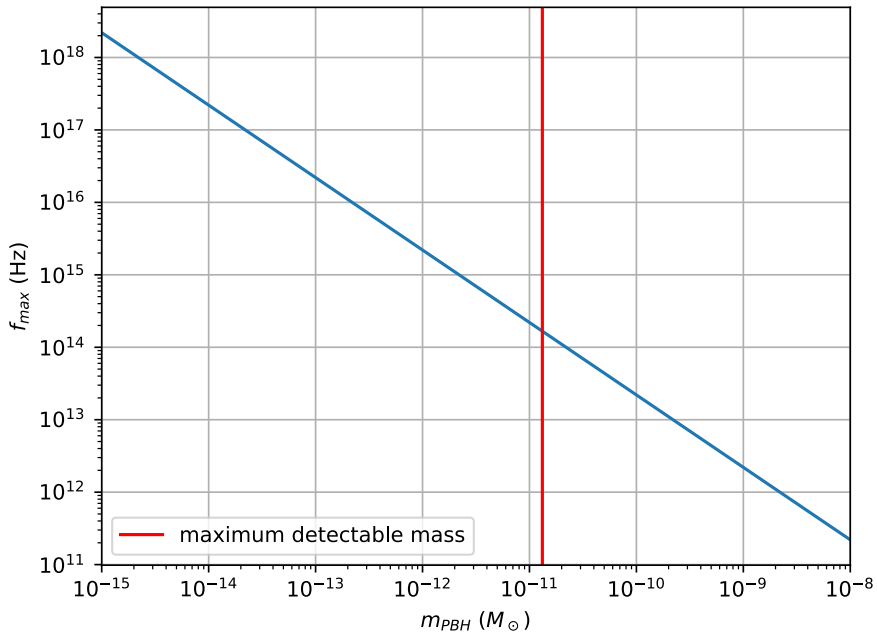


Figure 2.4: Maximum frequencies from Eq. (2.3.9) depending on the mass of equal mass black hole binaries. The highest detectable mass of $1.32 \cdot 10^{-11} M_\odot$ for the ALPS II experiment is marked with the red line.

Using the now time dependant expression for f_{gw} from Eq. (2.3.7) the strain of the gravitational waves (Eq. (2.3.2)) can be rewritten for the case of the slowly changing radius. For the time to coalescence τ as measured by the observer, the strain is given as

$$\begin{aligned} h_+(t) &= \frac{1}{r}\left(\frac{GM_c}{c^2}\right)^{5/4}\left(\frac{5}{c\tau}\right)^{1/4}\frac{1+\cos^2\theta}{2}\cos(\Phi(\tau)), \\ h_x(t) &= \frac{1}{r}\left(\frac{GM_c}{c^2}\right)^{5/4}\left(\frac{5}{c\tau}\right)^{1/4}\cos\theta\sin(\Phi(\tau)), \end{aligned} \tag{2.3.10}$$

with

$$\Phi(\tau) = -2\left(\frac{5GM_c}{c^3}\right)^{-5/8}\tau^{5/8} + \Phi_0. \tag{2.3.11}$$

Note here that Φ_0 is an integration constant equal to Φ at coalescence. Figure 2.5 shows the strain for PBHs of $10^{-11} M_\odot$ solar masses which is of the order of 10^{-25} at a distance of 1 pc.

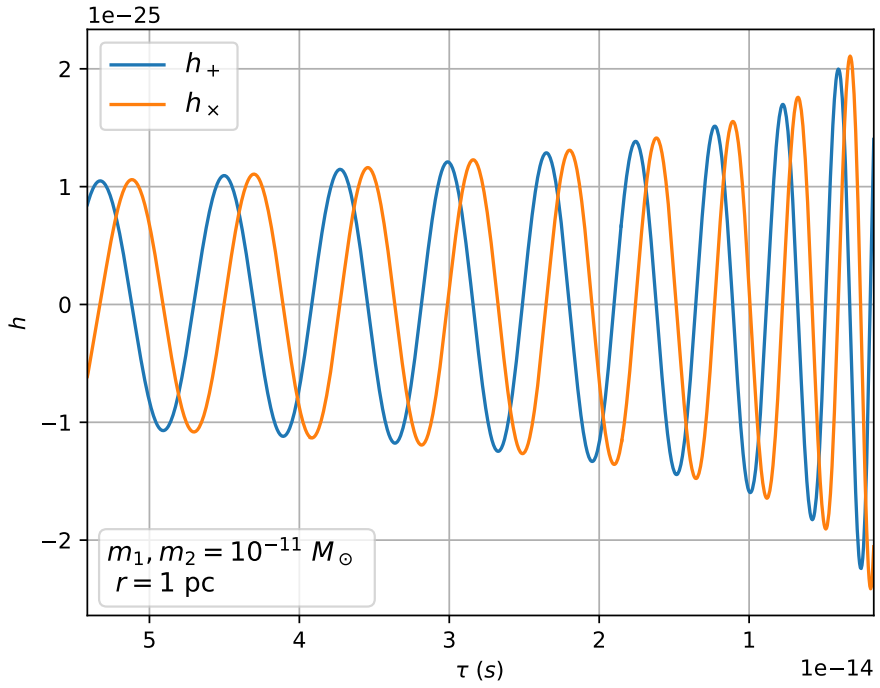


Figure 2.5: The strain for the gravitational waves from an equal mass PBH merger with $10^{-11} M_\odot$ as measured by an observer at a distance of 1 pc. The angle θ is set to 0 so that the observer is aligned with the binary rotation axis and $\Phi_0 = 0$.

For the detection of gravitational waves, one is generally less interested in the strain but more in its spectrum. Thus it is necessary to compute the Fourier transform of the strain given in Eq. (2.3.10) which is defined as:

$$\tilde{h}_{+,\times}(f) = \int dt\, h_{+,\times}(t) e^{i2\pi f t} \tag{2.3.12}$$

The Fourier transform of the strain is calculated in [18]:

$$\begin{aligned}
\tilde{h}_+(f) &= A e^{i\Psi_+(f)} \frac{c}{r} \left(\frac{GM_c}{c^3}\right)^{5/6} \frac{1}{f^{7/6}} \left(\frac{1+\cos^2\theta}{2}\right), \\
\tilde{h}_\times(f) &= A e^{i\Psi_\times(f)} \frac{c}{r} \left(\frac{GM_c}{c^3}\right)^{5/6} \frac{1}{f^{7/6}} \cos(\theta).
\end{aligned} \tag{2.3.13}$$

Here A is a numerical constant given as

$$A = \frac{1}{\pi^{2/3}} \left(\frac{5}{24}\right)^{1/2}. \tag{2.3.14}$$

The complex arguments are

$$\begin{aligned}
\Psi_+(f) &= 2\pi f(t_c + r/c) - \Phi_0 - \frac{\pi}{4} + \frac{3}{4}\left(\frac{GM_c}{c^3} 8\pi f\right)^{-5/3}, \\
\Psi_\times(f) &= \Psi_+(f) + \pi/2,
\end{aligned} \tag{2.3.15}$$

with the retarded time to coalesence $t_c = t_{coal} - r/c$. The Fourier transform of the strain from Eq. (2.3.13) will be important in the next Chapter to calculate how gravitational waves transform into a photon field within a magnetic field.

3 Graviton to Photon Conversion

To understand how it may be possible to detect gravitational waves in the ultra high frequency regime using experiments like the ALPS II experiment, first the process of graviton conversion needs to be analysed. This study follows closely the calculations made in Ref. [10] which looks at the the conversion of gravitational waves from the stochastic PBH background. It makes use of the graviton-photon mixing (GRAPH) which allows the conversion of gravitational waves to electromagnetic waves inside a magnetic field. Since the photon graviton mixing is calculated generally for a conversion inside a homogenous field it can thus also be used for gravitational waves from a binary merger source.

3.1 GRAPH in a Homogenous Magnetic Field

As is done in [10], rationalised Lorentz-Heaviside natural units with $k_B = \hbar = c = \epsilon_0 = 1$ will be used here. The starting point of computation is the total Lagrangian density composed of both the Lagrangian density for the electromagnetic and the gravitational field respectively:

$$\mathcal{L} = \mathcal{L}_{gr} + \mathcal{L}_{em} = \frac{1}{16\pi G}R - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}\int d^4x' A_\mu(x)\Pi^{\mu\nu}(x,x')A_\nu(x'). \tag{3.1.1}$$

Here, G is the Newtonian gravitational constant, R is the Ricci scalar, $F_{\mu\nu}$ is the electromagnetic field tensor and $\Pi^{\mu\nu}$ describes the photon polarisation tensor in a magnetic medium. Using the action $S = \int d^4x\sqrt{-g}\mathcal{L}$ (with the determinant of the metric g) it is possible to derive equations of motions for the electromagnetic and gravitational fields. This is done by expanding the metric tensor around the flat Minkowski metric $\eta_{\mu\nu}$ with the gravitational wave tensor $h_{\mu\nu}$:

$$g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu} + \dots, \tag{3.1.2}$$

with $\kappa^2 = 16\pi G$. The effective action following Ref. [11] is given as:

$$\begin{aligned} S_{\text{eff}} =& \frac{1}{4}\int d^4x[2\partial_\mu h^{\mu\nu}\partial_\rho h^\rho_{\ \nu} + \partial_\mu h\partial^\mu h - \partial_\mu h_{\alpha\beta}\partial^\mu h^{\alpha\beta} - 2\partial_\mu h^{\mu\nu}\partial_\nu h] \\ &- \frac{1}{4}\int d^4x F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}\int d^4x \int d^4x' A_\mu(x)\Pi^{\mu\nu}(x,x')A_\nu(x') \\ &+ \frac{\kappa}{2}\int d^4x h_{\mu\nu}T^{\mu\nu}_{\text{em}} + \mathcal{O}(\kappa\partial h^3) + \mathcal{O}(\kappa h\Pi), \end{aligned} \tag{3.1.3}$$

where $h = \eta_{\mu\nu}h^{\mu\nu}$ and $T^{\mu\nu}_{\text{em}}$ is the electromagnetic stress-energy tensor. Note that contrary to conventions $h_{\mu\nu}$ is not a dimensionless quantity but has the dimensions of energy. The considered case has gravitational waves propagating in a vacuum and then entering an external magnetic field. Therefore the electromagnetic field tensor is now split into the sum of the incident photon field tensor $f_{\mu\nu}$ and the external field tensor $\bar{F}_{\mu\nu}$:

$$F_{\mu\nu} = f_{\mu\nu} + \bar{F}_{\mu\nu} \tag{3.1.4}$$

Let A^μ be the four potential of the photon field and h_{ij} components of the tensor for the gravitational wave propagating inside the external magnetic field. Using the Euler-Lagrange equations and the action from Eq. (3.1.3) the resulting equations of motion (see Ref. [11]) are given by

$$\begin{aligned}
&\nabla^2 A^0 = 0, \\
&\Box \mathbf{A}^i + \left(\int d^4x' \Pi^{ij}(x, x') \mathbf{A}_j(x') \right) + \partial^i \partial_\mu A^\mu = \kappa \partial_\mu [h^{\mu\beta} \bar{F}^i_{\ \beta} - h^{i\beta} \bar{F}^\mu_{\ \beta}], \\
&\Box h_{ij} = -\kappa (B_i \bar{B}_j + \bar{B}_i B_j + \bar{B}_i \bar{B}_j).
\end{aligned} \tag{3.1.5}$$

For a non-varying magnetic field the equations can be linearised and simplified under the list of the following assumptions:

1. The gravitational wave travels in the $\hat{z}$ direction of a Cartesian coordinate system.
2. The external magnetic field is transverse to the photon direction of propagation spanning a finite length along the $\hat{z}$ direction and only has a component in $\hat{x}$-direction: $\bar{\mathbf{B}} = (\bar{B}_x, 0, 0)$.
3. The magnetic field is homogenous and there is a perfect vacuum inside the magnetic field.
4. Initially there is only a gravitational wave entering the magnetic field. Meaning that at $z = 0$ where the magnetic field begins the electric field is zero.

The Fourier modes $\tilde{A}_x$ and $\tilde{A}_y$ of the electromagnetic four potential at a distance z inside the magnetic field are calculated in [10]. For the total energy of the fields ω they depend on the Fourier modes of the incoming gravitational wave $\tilde{h}_\times(0, \omega, \hat{z}), \tilde{h}_+(0, \omega, \hat{z})$ at $z = 0$:

$$\begin{aligned}
\tilde{A}_x(z, \omega, \hat{z}) &= -\frac{M^x_{g\gamma} \sin(\Delta_x z)}{4\sqrt{\pi G}\Delta_x} e^{i(\omega + M_x/2)z} \tilde{h}_\times(0, \omega, \hat{z}), \\
\tilde{A}_y(z, \omega, \hat{z}) &= \frac{M^x_{g\gamma} \sin(\Delta_y z)}{4\sqrt{\pi G}\Delta_y} e^{i(\omega + M_y/2)z} \tilde{h}_+(0, \omega, \hat{z}).
\end{aligned} \tag{3.1.6}$$

Here $\Delta_{x,y}$ is defined as

$$\Delta_{x,y} = 1/2 \cdot \sqrt{M^2_{x,y} + 4(M^x_{g\gamma})^2}, \tag{3.1.7}$$

with the expressions for the elements of the mass mixing matrix $M^x_{g\gamma}, M_x$ and M_y derived in Ref. [11]. More precisely $M^x_{g\gamma} = 2\sqrt{\pi G}\bar{B}_x$ and $M_{x,y}$ depend on the photon polarisation tensor and are both zero for the case of pure vacuum in the magnetic field. Thus $\Delta_{x,y} = M^x_{g\gamma}$ and Eq. (3.1.6) simplifies to:

$$\begin{aligned}
\tilde{A}_x(z, \omega, \hat{z}) &= -\frac{\sin\left(M^x_{g\gamma} z\right)}{4\sqrt{\pi G}} e^{i\omega z} \tilde{h}_\times(0, \omega, \hat{z}), \\
\tilde{A}_y(z, \omega, \hat{z}) &= \frac{\sin\left(M^x_{g\gamma} z\right)}{4\sqrt{\pi G}} e^{i\omega z} \tilde{h}_+(0, \omega, \hat{z}).
\end{aligned} \tag{3.1.8}$$

Using the Fourier transform the elements $A_x, y(z,t)$ of the four potential can be expressed as:

$$A_{x,y}(z,t) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \tilde{A}_{x,y}(z,\omega) e^{-i\omega t}. \tag{3.1.9}$$

Note that the direction $\hat{z}$ is left out for convenience from here on but is still implied. The electric field $E_{x,y}(z,t)$ is related to the four potential through

$$E_{x,y}(z,t) = -\nabla A^0 - \partial_t A(x,t). \tag{3.1.10}$$

In vacuum the electric potential $A^0 = 0$ and thus $E_{x,y}(z,t) = -\partial_t A(x,t)$. Using this together with Eq. (3.1.9) results in the following expressions for the electric field components:

$$E_{x,y}(z,t) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} i\omega \tilde{A}_{x,y}(z,\omega) e^{-i\omega t} \tag{3.1.11}$$

3.2 Energy Flux of The Electromagnetic Radiation

To determine the signal flux observed at a detector it is vital to first compute the energy flux as in the end an intensity is measured in the experiment. The energy density can be computed through the Stokes parameter:

$$\Phi_\gamma(z,t) = \langle |E_x(z,t)|^2 \rangle + \langle |E_y(z,t)|^2 \rangle. \tag{3.2.1}$$

Note that $\langle |E_{x,y}(z,t)|^2 \rangle$ denotes the temporal average over many oscillation periods for $|E_{x,y}(z,t)|^2$ which can be expressed explicitly using a double integral (see Eq. (3.1.11)):

$$|E_{x,y}(z,t)|^2 = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} \omega\omega' \tilde{A}_{x,y}(z,\omega) \tilde{A}^*_{x,y}(z,\omega') e^{i(\omega'-\omega)t}. \tag{3.2.2}$$

Assuming an observation time T starting at $t = t_0$ the temporal average is given as:

$$\langle |E_{x,y}(z,t)|^2 \rangle = \frac{1}{T} \int_{t_0}^{t_0+T} dt \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} \omega\omega' \tilde{A}_{x,y}(0,\omega) \tilde{A}^*_{x,y}(0,\omega') e^{i(\omega'-\omega)t}. \tag{3.2.3}$$

To simplify Eq.(3.2.3) it is of use to consider the integral over time explicitly and compute the Fourier transform of the rectangular function.

$$\begin{aligned} \int_{t_0}^{t_0+T} e^{i(\omega'-\omega)t} dt &= \int_{-\infty}^{\infty} \mathrm{rect}\left(\frac{t-(t_0+T/2)}{T}\right) e^{-i(\omega-\omega')t} dt \\ &= e^{-i(\omega-\omega')(t_0+T/2)} \int_{-\infty}^{\infty} \mathrm{rect}\left(\frac{t}{T}\right) e^{-i(\omega-\omega')t} dt \\ &= T\,\mathrm{sinc}\left(T \cdot \frac{\omega-\omega'}{2}\right) e^{i(\omega'-\omega)(t_0+\frac{T}{2})}. \end{aligned} \tag{3.2.4}$$

The unnormalised sinc function in Eq. (3.2.4) is defined as

$$\mathrm{sinc}(x) = \begin{cases} \frac{\sin(x)}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}. \tag{3.2.5}$$

With Eq. (3.2.4) the integral over time in Eq. (3.2.3) can be eliminated:

$$\langle |E_{x,y}(z,t)|^2 \rangle = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} \omega\omega' \tilde{A}_{x,y}(z,\omega) \tilde{A}^*_{x,y}(z,\omega') \mathrm{sinc}\left(T \cdot \frac{\omega - \omega'}{2}\right) e^{i(\omega'-\omega)(t_0+\frac{T}{2})}. \tag{3.2.6}$$

Assuming the time of observation T is sufficiently large the sinc function can be approximated as a δ-distribution:

$$\delta(x) = \lim_{T\to\infty} \frac{\sin(xT)}{\pi x} = \lim_{T\to\infty} T \frac{\mathrm{sinc}(xT)}{\pi}. \tag{3.2.7}$$

Introducing this into Eq. (3.2.6) yields

$$\langle |E_{x,y}(z,t)|^2 \rangle = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi T} \int_{-\infty}^{\infty} d\omega' \omega\omega' \tilde{A}_{x,y}(z,\omega) \tilde{A}^*_{x,y}(z,\omega') \delta(\omega - \omega') e^{i(\omega'-\omega)(t_0+\frac{T}{2})}, \tag{3.2.8}$$

or explicitly

$$\langle |E_{x,y}(z,t)|^2 \rangle = \frac{1}{2\pi T} \int_{-\infty}^{\infty} \omega^2 |\tilde{A}_{x,y}(z,\omega)|^2 d\omega. \tag{3.2.9}$$

The Stokes parameter from Eq. (3.2.1) is therefore given as:

$$\Phi_\gamma(z,t) = \frac{1}{2\pi T} \int_{-\infty}^{\infty} \omega^2 \left(|\tilde{A}_x(z,\omega)|^2 + |\tilde{A}_y(z,\omega)|^2 \right) d\omega. \tag{3.2.10}$$

Substituting the expression for $\tilde{A}_{x,y}(z,\omega)$ given in Eq. (3.1.8) the energy density of the photon field depends on the gravitational wave strain as follows:

$$\Phi_\gamma(z,t) = \frac{\sin^2(M^x_{g\gamma} z)}{32\pi^2 GT} \int_{-\infty}^{\infty} \omega^2 \left(|\tilde{h}_\times(0,\omega)|^2 + |\tilde{h}_+(0,\omega)|^2 \right) d\omega. \tag{3.2.11}$$

The integral over the total energy ω can also be expressed as an integral over the frequency $f = \frac{\omega}{2\pi}$:

$$\Phi_\gamma(z,t) = \frac{\pi \sin^2(M^x_{g\gamma} z)}{4GT} \int_{-\infty}^{\infty} f^2 \left(|\tilde{h}_\times(0,f)|^2 + |\tilde{h}_+(0,f)|^2 \right) df. \tag{3.2.12}$$

Through dimensional analysis Eq. (3.2.12) can be expressed in SI units by reintroducing factors of c and ϵ_0. Simultaneously replacing $M^x_{g\gamma} = 2\sqrt{\pi G}\bar{B}_x$ leads to:

$$\Phi_\gamma(z,t) = \frac{\pi c^2}{4\epsilon_0 GT} \sin^2\left(\frac{2\sqrt{\pi\epsilon_0 G}\bar{B}_x z}{c}\right) \int_{-\infty}^{\infty} f^2 \left(|\tilde{h}_\times(0,f)|^2 + |\tilde{h}_+(0,f)|^2 \right) df. \tag{3.2.13}$$

This expression together with the gravitational wave strain for a binary merger as calculated in Chapter 2 will be used in Chapter 5 to determine the possible signal strength with the detectors at ALPS II .

4 Single Photon Detection

In the previous Chapter 3 the photon signal a gravitational wave could generate inside a magnetic field has been derived. Detection of such a weak signal requires the use of highly sensitive and low noise detectors. Two of these detectors will be discussed here: Firstly the Transition Edge Sensor (TES) being a main focus of research for the ALPS II experiment. Secondly the Superconducting Nanowire Single-Photon Detector (SNSPD) which is currently being tested for efficiency at the University of Southern Denmark (Section 4.3). Both of these sensors need to be operated at extremely low temperatures as they make use of superconductive materials. In section 4.4 a simple overview of the refrigerator used to cool the SNSPD in the Dark Quantum Lab at SDU is showcased. A short introduction to superconductivity will be given in Section 4.1 taking inspiration from the textbook "Festkörperphysik" by Gross and Marx [14].

4.1 Superconductivity

In 1911 Heike Kamerlingh Onnes observed that at very low temperatures the resistance of mercury dropped suddenly to an immeasurable small amount [27]. At the temperature of 4.2 K mercury becomes superconductive. Since this groundbreaking discovery superconductivity has come a long way and finds itself nowadays in a variety of different technical applications in science and industry.

Superconducting materials are characterised by a critical temperature T_c, a critical magnetic field B_c and a critical current density J_c, below which they transition into a superconductive state [14]. In this state the resistivity is zero and any outside magnetic field is expelled from the interior of the superconducting lattice (Meißner-Ochsenfeld effect). Furthermore T_c, B_c and J_c all depend on each other. The zero resistance results in the absence of power dissipation in superconducting circuits, making superconductors the enabling technology for high-field magnets and quantum computing, just to name a few (see for example Ref. [28]).

Superconductivity can be explained through the BCS-theory named after J. Bardeen, L.N. Cooper and J.R. Schrieffer [5]. The core aspect of BCS-theory is the creation of Cooper pairs. These are pairs of electrons with antiparallel spin and momentum that are bound by attractive forces created through interactions with the ionic lattice. While this is a quantum effect it can be understood classically as the electrons distorting the ionic lattice around them and attracting surrounding ions. This creates a phonon (quanta lattice vibration) mediated attractive force between the electrons resulting in the formation of Cooper pairs.

A Cooper pair can be treated as a single particle with spin zero corresponding to a boson. So contrary to fermionic, non bound electrons Cooper pairs do not underlie Fermi's exclusion principle. Therefore, as bosons they can condense into the same ground state which can be described by a single wave function. A key result of BCS-theory is that the condensation of Cooper pairs results in the formation of an energy gap between the condensate ground state and the first excited state. This energy gap prevents electron scattering that would result in power dissipation in normal metals, thus giving rise to

superconductivity.

The phase transition between superconducting and normal conductivity is characterised by a very steep slope in the resistance at T_c. At this point that is called the transition edge, the resistance is highly temperature sensitive as can be seen in Figure 4.1 for an example superconductor. This makes the transition region highly useful as a sensitive thermometer and can be used in sensors like the TES as described in Section 4.2.

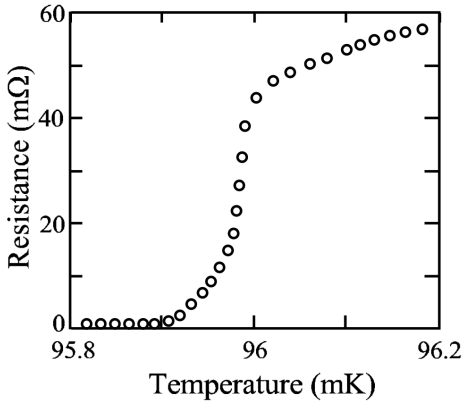


Figure 4.1: Resistance of a superconducting film made of a Mo/Cu proximity bilayer in its superconductive phase transition around 96 mK. The image is taken from Ref. [16].

Superconductors can be classified into two types: For type I superconductors once B_c is reached the magnetic field on the interior is no longer expelled and superconductivity breaks down. The critical field is temperature dependant and is at its highest at absolute zero:

$$B_c(T) = B_c(0)\left[1 - \left(\frac{T}{T_c}\right)^2\right]. \tag{4.1.1}$$

Type-II superconductors on the other hand are characterised by two critical magnetic field values B_{c1} and B_{c2}. Below B_{c1} the superconductor is in the Meissner state where the magnetic field is expelled. Between B_{c1} and B_{c2} it finds itself in a mixed states with the magnetic field entering the superconductor in the form of vortices (quanta of magnetic flux). Superconductivity does not break down however until B_{c2} is reached.

4.2 TES

At ALPS II the preparations for implementing a Transition Edge Sensor (TES) for single photon detection are still ongoing, the status of which can be seen in Refs. [12],[24]. The thin-film tungsten based TES is to be integrated in parallel with the currently used heterodyne based detection scheme. It is operated at a temperature close to the transition temperature to its superconducting phase of around 140 mK [12]. The TES has many properties making it a highly sensitive single photon detector (see for example Refs. [8],[16]). For starters it has a good energy resolution (5.3% according to Ref. [12]) and has a high quantum efficiency. Additionally it has a very low dark count rate and only a very short dead time in the microsecond range. In Ref. [9] the TES working

principle as well as its operation are described in detail. A more compact overview will be given in Section 4.2.1.

4.2.1 TES Working Principle

The main setup of the TES involves an electrical circuit and a thermal circuit. A schematic is shown in Figure 4.2 taken from [9]. Note that the electrical circuit shown does not represent the usual technical implementation but an equivalent version that can be used for explanation. The realisation of an actual TES circuit can be found in Ref. [16].

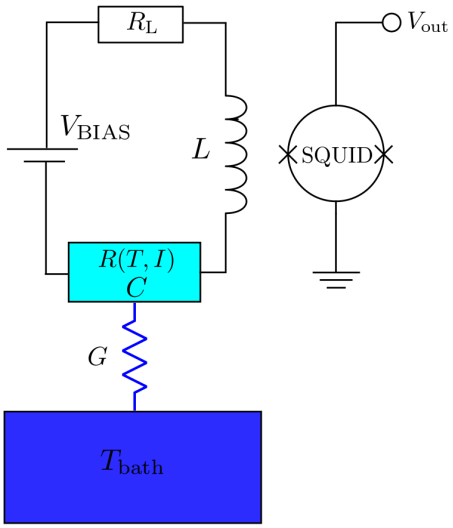


Figure 4.2: Thermal and electrical system of the TES from [9]. The sensor itself is shown in cyan and has the heat capacity C and electric resistance $R(T, I)$ depending on both the temperature and the current. It is in contact with a thermal bath at a temperature of T_{bath} (dark blue) and is connected via a link with thermal conductance G. The sensor is incorporated in an electrical circuit supplied by a constant bias voltage V_{BIAS}. The circuit also includes a load resistance R_L and an inductance L. The inductance is coupled to a SQUID which reads out the current and gives an output voltage V_{out} proportional to set current.

To be able to use the TES as a sensitive thermometer the sensor needs to be operated in the transition edge. At this working point with temperature T_0 the sensor is in the state between superconducting and normal conduction. As seen in Figure 4.2 the sensor itself is connected to a thermal bath with temperature T_{bath}. Connecting them is a thermal link with conductance G. This allows the sensor to cool back down to the working point after heating up through for example the absorption of a photon. The temperature difference ΔT the sensor experiences depends on its conductance C and the absorbed energy ΔE:

$$\Delta T = \frac{\Delta E}{C}. \tag{4.2.1}$$

At this point through heat transfer the temperature of the sensor decreases back to the temperature of the thermal bath it is in contact with. The drop in temperature over the time t is described by an exponential decay

$$T(t) \sim e^{-t/\tau} \tag{4.2.2}$$

with the thermal time constant $\tau = C/G$. In theory it would be possible to set the working point of the TES through the temperature of the bath alone. However, in practice it is fairly difficult to achieve the slim transition window just by regulating the temperature through T_{bath}. Therefore T_{bath} is set to below T_c where the sensor is fully superconducting. Now a current is set in the electrical circuit and varied until the critical current density J_c is reached which starts breaking superconductivity. The voltage needed to set the sensor at the working point in the superconducting transition is the constant bias voltage V_{BIAS}. For small signals the resistance can be expended around R_0, T_0 and I_0 which are the resistance, temperature and current at the working point respectively:

$$R(T, I) \approx R_0 + \left.\frac{\partial R}{\partial T}\right|_{I_0} \delta T + \left.\frac{\partial R}{\partial I}\right|_{T_0} \delta I. \tag{4.2.3}$$

Here $\delta T = T - T_0$ is a small change in temperature and $\delta I = I - I_0$ is a small change in current. Since at the working point the resistance R_0 is specifically non zero the Joule power of the electrical circuit $P_{J_0} = V_{\text{BIAS}}^2/R_0$ heats up the system. This heating gets absorbed into the power flow to the bath P_{bath} keeping the system in a fairly stable state. That is until the sensor heats up through an external energy. This causes an increase in the resistance and therefore a lower current in the system. As a result the heating power decreases and the sensor returns to its working temperature after a certain time depending on the amount of initially deposited energy.

The TES therefore creates a negative electrothermal feedback (negative ETF): An increase in temperature of the sensor results in a lower current stabilising the system and bringing it back to base temperature. Negative ETF is therefore a consequence of the constant bias voltage that is being applied. In contrast, applying a constant current instead of voltage would create a TES with positive ETF. Operating the TES with a bias voltage has a couple of advantages though like for example faster signals and no risk of thermal runaway.

So to summarise when a photon is absorbed by the sensor the deposited energy causes a rise in temperature which increases the resistance. Because of the constant bias voltage the current in the circuit decreases. The change in current can be measured by an inductively coupled Superconducting Superconducting Quantum Interference Device (SQUID). The SQUID is a highly sensitive magnetometer formed by a superconducting loop interrupted by two Josephson junctions. These junctions can be realised by placing a thin isolating layer between superconducting materials. Between the superconductors a supercurrent can flow as the result of the tunnelling of Cooper pairs which is the so called Josesphson effect [14].

The SQUID is characterised by a sinusoidally changing voltage as a function of magnetic flux penetrating through the SQUID loop. Thus it can be used to detect extremely small changes in external magnetic fields, such as the change in magnetic field due to the change of the TES current. As a result the change in the output voltage of the SQUID V_{out} is proportional to the change of the current in the TES.

4.2.2 TES Reflectance

To determine the possible photon signal the TES sensor could detect at ALPS II certain characteristics of the TES module need to be known. An in depth analysis of the TES

module can be found in Ref. [9]. The most important result concerning this study is the reflectance which is presented here.

Firstly the TES module is based on a 20 nm tungsten film fabricated by the U.S. National Institute of Standards and Technology (NIST). The detection efficiency is optimised for 1064 nm photons at which point it is nearly at unity. To maximise the absorption spectrum for 1064 nm photons the TES is incorporated into an optical stack. The reflectance of the TES for different wavelengths of incoming light are shown in Figure 4.3. A model of this reflectance will be used in Chapter 5 to estimate the efficiency of the TES over a bandwidth of different frequencies.

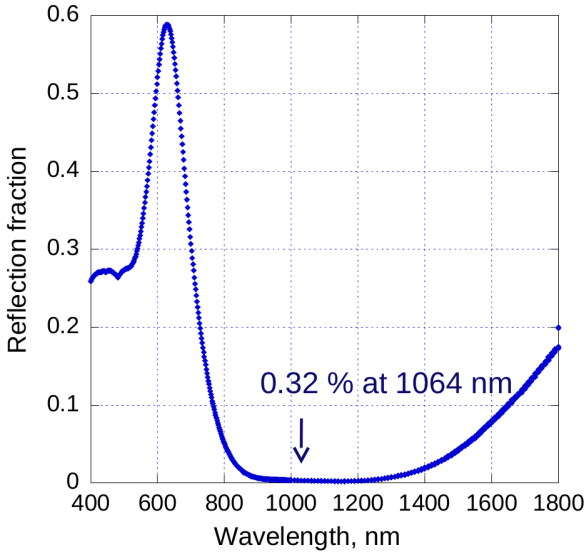


Figure 4.3: Reflectance of the NIST TES module for different wavelengths as measured in the warm taken from Ref. [9]. At the optimised wavelength of 1064 nm the reflectance is only at 0.32 %.

While the efficiency of the TES is very high for the optimised wavelengths for signal significance also the rate of background events needs to be considered. This will be done in Chapter 5 when the actual photon signal for gravitational waves is calculated.

4.3 SNSPD

Superconducting Nanowire Single-Photon Detectors (SNSPDs) are in many aspects similar to TESs as both of them rely on biasing the superconductor close to the superconducting phase transition and photon detection results from the heat-up of the superconductor during photon absorption. The detector element of the SNSPDs consists of a patterned wire on a nm-scale that is current-biased close to its superconducting critical current (typically around 90% of I_c). The absorption of a photon heats up the entire width of the nanowire breaking superconductivity. This results in an increase in resistance (in the kOhm range) within a very short time frame, typically measured in picoseconds. The readout of the SNSPD differs significantly from the TES introduced in Section 4.2. Instead of coupling the nanowire to a SQUID, it uses a bias-tee, the equivalent circuit of which is illustrated in Figure 4.4 (see for example Ref [17]).

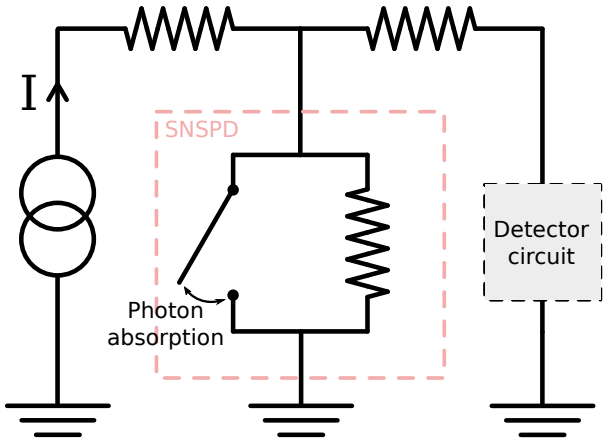


Figure 4.4: Schematic of an equivalent circuit of the bias-tee to read out the SNSPD. Absorption of a photon corresponds to the opening of a switch in the SNSPD circuit.

While the nanowire is superconducting, most of the applied current will flow through it to ground. When the absorption of a photon turns the nanowire into its resistive state, the current flows into the detection circuit instead, where the associated voltage pulse can be directly measured, e.g. with a high-time-resolution oscilloscope. Arguably, the main advantage of SNSPDs when compared with TESs is their simple design and low-cost. This results from the possibility to fabricate amorphous and polycrystalline superconducting thin films using room temperature sputtering, which has enabled the efficient and low-cost production of SNSPDs that are nowadays widely commercially available [17]. Moreover, these materials, such as NbN and NbTiN, are type-II superconductors which are associated with significantly higher superconducting transition temperatures when compared to the commonly tungsten based TESs [8].

As a consequence, while a TES requires a dilution fridge to be brought to operation temperature, SNSPDs can rely on less expensive refrigeration systems. Another advantage of utilizing type-II superconductors is that SNSPDs have extremely high critical magnetic fields. This can be a great benefit for various axion experiments. Another advantage of SNSPDs is their short dead times in the scale of nanoseconds [15]. This results in significantly higher maximum count rates when compared with TESs, whose dead times are typically in the μs range [8].

One of the downsides of SNSPDs is that they have very poor energy resolution when compared to the TES. Therefore they are often only used as binary detectors. However, this can be overcome by applying an optical bandpass filter in front of the SNSPD resulting in effective energy resolution equal to the transmission window of the used filter. Additionally, the detection efficiency of SNSPDs, particularly for photons in the near-infrared range, have been historically significantly lower when compared to the high quantum efficiency of TESs. Recent developments in the field have, however, resulted in major improvements and state of the art SNSPDs have detection efficiencies comparable to those of TESs [19],[29].

4.4 Refrigeration of superconducting detectors

Most of the refrigeration systems today rely on the use of pulse tube cryocoolers that can cool down below 4 K temperature via periodic compression and expansion of a high-

pressure working gas (typically helium). This means that the pulse tubes can be used for liquefaction of helium, which can be then further pumped in order to reach even lower temperatures. The cryostat used to operate SNSPDs in the Dark Quantum Lab at SDU operates according to this principle. A schematic illustration of this dry closed-loop He-4 circulation refrigerator is presented in Figure 4.5.

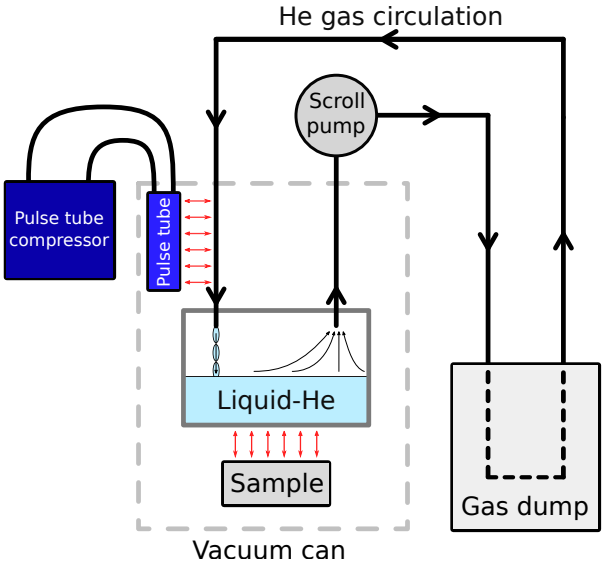


Figure 4.5: Schematic illustration of the cryostat in the Dark Quantum Lab at SDU. The Pulse tube cools down the circulating helium gas that liquefies inside the sample space inside a vacuum can where it cools the sample. A scroll pump causes the liquid helium to evaporate causing further cooling and allows the gas to flow back into the Gas dump.

The He gas is pumped into the vacuum can where it comes into good thermal contact with the pulse tube, which results in the condensation of the helium. At this point the liquid-He bath that has formed is pumped resulting in efficient evaporative cooling ultimately lowering the temperature of the sample space down to 1.5 K. The evaporated He is then returned back in the gas dump where it is circulated back to the Pulse tube and the cycle can continue. While this simple cryostat design is sufficient for the operation of SNSPDs, it cannot reach below 1 K temperatures required for the operation of TESs. These are operated in more sophisticated dilution fridges, which use He-3/He-4 mixtures as their working gas to provide cooling at 10 mK temperatures. More information about cryogenics and refrigerators at low temperatures can be found for example in [22].

5 Gravitational Wave Detection with ALPS II

In this chapter the main results of the study will be presented. To this end a short introduction to the ALPS II experiment will be given in Section 5.1. Subsequently in Section 5.2 the theoretical photon signal of a PBH binary merger as detected with the ALPS II setup is computed. Using background rates for the TES detector from Ref. [20] the significance of set signal is discussed.

5.1 The ALPS II Experiment

The ALPS II experiment is the successor of the Any Light Particle Search (ALPS) which took place from 2007 to 2010 at DESY ("Deutsches Elektronen-Synchrotron ") in Hamburg. It is a so called "Light Shining through Wall" (LSW) experiment searching for axions or axion like particles (ALPs). These are hypothetical weakly interacting particles and are a prime dark matter candidate. Inside of a magnetic field ALPs can convert into photons through ALP-photon mixing [23].

The general setup for ALPS II is shown in Figure 5.1 from [2]. A Laser with a wavelength of 1064 nm shines into the Production Cavity (PC). Inside this cavity 10 superconducting dipole magnets create a magnetic field of 5.3 T over a length of 105.6 m [24]. Optical resonators amplify the photon field inside the cavity to increase the probability of ALP production. The laser power is boosted by a build-up factor of $\beta_{PC} = 5000$ [21]. If an ALP is produced it can pass through an opaque wall into the Regeneration Cavity (RC). This cavity has an identical array of magnets as the Production Cavity but uses a different set up of optical resonators to gain a power build-up of $\beta_{RC} = 40000$ [21]. Inside the magnetic field the ALPs can convert back to photons which are guided to a detector. The measurement of a photon signal behind the opaque wall would thus prove the existence of ALPs. For the first data taking run a heterodyne detection scheme was used [21]. In the future the TES will be implemented as a secondary detection method to possibly provide better performance.

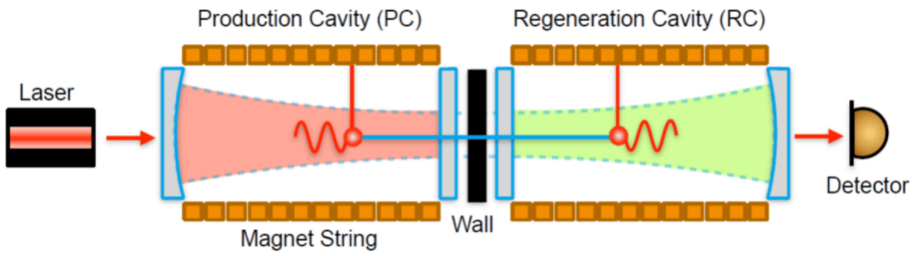


Figure 5.1: Schematic of the ALPS II setup taken from [2]. The Laser shines light into the Production Cavity (PC) where a photon can convert into an ALP inside the magnetic field. The ALP can pass through the wall and reconvert into a photon inside the Regeneration Cavity (RC). A detector measures the photon field behind the regeneration cavity. Optical resonators enhance the photon field in both cavities.

From the setup it becomes clear that the ALPS II experiment could theoretically lend itself to the detection of gravitational waves. In this case only the Regeneration Cavity is of interest. Over the length of 105.6 m a gravitational wave signal can create a photon signal inside the 5.3 T strong magnetic field. In the following Section 5.2 the signal that could be detected by the TES is analysed.

5.2 Gravitational Wave Signal at ALPS II

Now that the setup of ALPS II is clear it is time to compute the gravitational wave signal as detected by the TES. The Fourier transform of the strain for a binary merger at distance r is given in Eq. (2.3.13). Assuming the magnetic field of ALPS II is oriented so it is transverse to the arriving gravitational wave the graviton photon mixing is at a maximum. It is unlikely that a random merger is positioned just so that the gravitational wave enters the magnetic field exactly perpendicular. However since the question is whether a merger might be detected at all it is sensible to consider the optimal case first. If the gravitational wave enters the magnetic field at an angle the calculations for the GRAPH mixing in Chapter 3 would need to be adjusted accordingly. For the maximum case using Eq. (3.2.13) and inserting the strain from Eq. (2.3.13) yields:

$$\langle |E_x(z,t)|^2 \rangle + \langle |E_y(z,t)|^2 \rangle = \frac{5G^{2/3}M_c^{5/3}}{96\pi^{1/3}c\epsilon_0 T r^2} \sin^2\left(\frac{2\sqrt{\pi\epsilon_0}G\bar{B}_x z}{c}\right) g(\theta) \int_{-\infty}^{\infty} f^{-1/3} df. \tag{5.2.1}$$

Here $g(\theta)$ is a function that includes the entire dependence on the binary orientation angle:

$$g(\theta) = \left(\frac{1+\cos^2\theta}{2}\right)^2 + \cos^2\theta. \tag{5.2.2}$$

The angular dependence $g(\theta)$ is shown in Figure 5.2. The function $g(\theta)$ is at maximum for an angle of $\theta = 0$ which is when the binary rotation axis is aligned with the line of sight of the observer: $g(0) = 2$. The weakest signal will be observed for a binary rotation angle of $\pi/2$ where $g(\pi/2) = 0$.

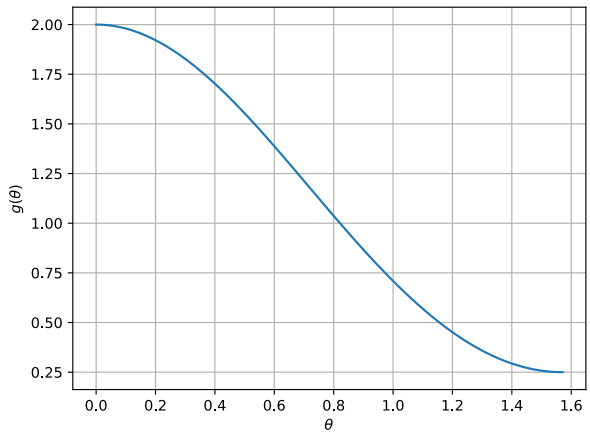


Figure 5.2: Signal dependence on the binary rotation angle $g(\theta)$ for $\theta \in [0, \pi/2]$ (see Eq. (5.2.2)).

For an electromagnetic wave in vacuum the time averaged intensity is given as

$$I_{\text{avg}} = \epsilon_0 c \langle \mathbf{E}^2 \rangle. \tag{5.2.3}$$

To get to the energy E that passes through the area A in a time Δt the intensity needs to be multiplied by both:

$$E = I_{\text{avg}} A \Delta t. \tag{5.2.4}$$

In the case of ALPS II the considered area A is the cross-sectional area of the beam pipe which has a diameter of 55.3 mm [2]. This results in an area of $A = 2.4 \cdot 10^{-3}\text{m}^2$. Therefore over the entire observation time ($\Delta t = T$) the energy can be expressed as

$$E = \frac{5AG^{2/3}M_c^{5/3}}{96\pi^{1/3}r^2} \sin^2\left(\frac{2\sqrt{\pi\epsilon_0 G}\bar{B}_x z}{c}\right) g(\theta) \int_{-\infty}^{\infty} f^{-1/3} df. \tag{5.2.5}$$

Concerning the observation time T it is important to note that in Chapter 3 for the computation of the Stokes parameter it was assumed T was sufficiently large, allowing the sinc function to be replaced by the δ-distribution. This is a very reasonable assumption since the usual observation time spans a length of several hours or days. However, already for very small observation times far below 1 s numerical computations showed that this is an adequate assumption.

Since the TES is detecting single photons it is important to determine the total amount of photons N_γ that are produced. To do this the total energy E which is given as an integral over the frequency must be divided by the energy of a single photon $E_\gamma = hf$:

$$N_\gamma = \frac{5AG^{2/3}M_c^{5/3}}{96h\pi^{1/3}r^2} \sin^2\left(\frac{2\sqrt{\pi\epsilon_0 G}\bar{B}_x z}{c}\right) g(\theta) \int_{-\infty}^{\infty} f^{-4/3} df. \tag{5.2.6}$$

At this point it is necessary to consider the efficiency of the TES. The quantum efficiency is determined by the absorbance η_{TES} of the TES. which is related to the reflection fraction defined as $1 - \eta_{\text{TES}}$. For the purpose of computation the reflection fraction from [9] shown in Figure 4.3 will need to be modelled. This is done following Ref. [25] by approximating the function using two error functions:

$$\begin{aligned} \eta_{\text{TES}}(\lambda) &= f_1(\lambda) \cdot f_2(\lambda) \\ f_1(\lambda) &= \frac{1-b_1}{2}(1 + \text{erf}\{a_1(\lambda - \lambda_1)\}) + b_1 \\ f_2(\lambda) &= \frac{1-b_2}{2}(1 - \text{erf}\{a_2(\lambda - \lambda_2)\}) + b_2. \end{aligned} \tag{5.2.7}$$

In accordance with Ref. [25] the parameters $a_1, a_2, b_1, b_2, \lambda_1$ and λ_2 are chosen so the modelled reflection fraction resembles the reflection data from [9] (Figure 4.3) as closely as possible. The modelled reflection fraction for the optimal parameters is shown in Figure 5.3. Comparing to the data the model describes the curve well for the larger wavelengths. It does underestimate the peak at around 600 nm and overestimates the tail at around 400 nm. This difference should not be significant for the outcome of this analysis though.

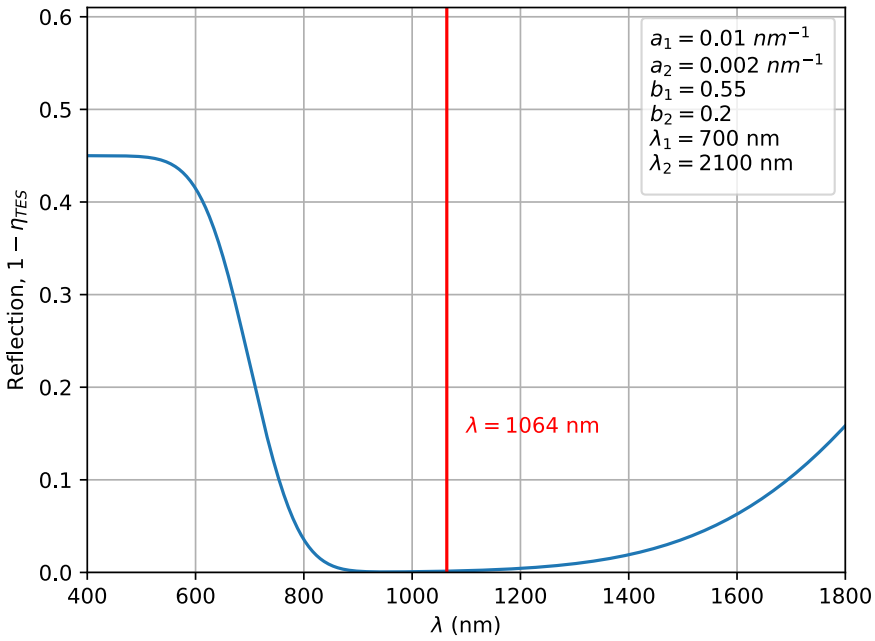


Figure 5.3: Model of the TES reflectance as a function of the wavelength λ from Eq. (5.2.7). As is done in Ref. [25] the parameters $a_1, a_2, b_1, b_2, \lambda_1$ and λ_2 are chosen so the modelled reflection curve resembles the reflection data from Ref. [9] shown in Figure 4.3.

Next to the quantum efficiency of the TES there is also other challenges with the setup of the TES detection scheme that can have an impact on the total detector efficiency. This comes mainly from the fact that the photon signal is guided to the TES via optical fibres. The optical efficiency ϵ_{opt} can be estimated as the product of the efficiency from the fiber-to-TES coupling (95%), the fiber transmittance (60%) and the beam-to-fiber coupling (80%) [9]. The number of photons absorbed by the TES $N_{\gamma,\text{TES}}$ includes thus both the absorption η_{TES} as a function of the frequency and the optical efficiency $\epsilon_{\text{opt}} = 0.46$:

$$N_{\gamma,\text{TES}} = \frac{5AG^{2/3}M_c^{5/3}}{96h\pi^{1/3}r^2} \sin^2\left(\frac{2\sqrt{\pi\epsilon_0 G}\bar{B}_x z}{c}\right) g(\theta)\epsilon_{\text{opt}} \int_{-\infty}^{\infty} \eta_{\text{TES}}(f) f^{-4/3} df. \tag{5.2.8}$$

Since the efficiency is only known in the range of $400 - 1800$ nm corresponding to frequencies of $1.67 - 7.49 \cdot 10^{14}$ Hz using the modelled η_{TES} for all frequencies would prove faulty. Therefore the integral will be calculated only for the frequency regime in which the efficiency is known:

$$N_{\gamma,\text{TES}} = \frac{5AG^{2/3}M_c^{5/3}}{96h\pi^{1/3}r^2} \sin^2\left(\frac{2\sqrt{\pi\epsilon_0 G}\bar{B}_x z}{c}\right) g(\theta)\epsilon_{\text{opt}} \int_{f_1}^{f_2} \eta_{\text{TES}}(f) f^{-4/3} df, \tag{5.2.9}$$

with $f_1 = 1.67$ Hz, $f_1 = 7.49 \cdot 10^{14}$ Hz. This will underestimate the signal but since the TES is so optimised for 1064 nm the efficiency should decrease fairly quickly for wavelengths $\lambda \ll 1064$ nm or $\lambda \gg 1064$ nm.

It is important to note that since the gravitational wave signal is computed for the inspiral phase of the PBH merger f_2 can not exceed the maximum frequency f_{max} from Eq. (2.3.9) for the innermost stable circular orbit. This concerns the PBH with masses close to the maximum detectable mass of $1.32 \cdot 10^{-11} M_\odot$. For the computation of their

signal $f_2 = f_{\max}$. For three equal mass binaries $N_{\gamma,\mathrm{TES}}$ as a function of the distance to the observer r from Eq. (5.2.9) is shown in Figure 5.4. In accordance with the ALPS II setup $B_x = 5.3$ T, $z = 105.6$ m and $A = 2.4 \cdot 10^{-13}$ m^2. To maximise the signal the binary orientation angle of the mergers is set to zero.

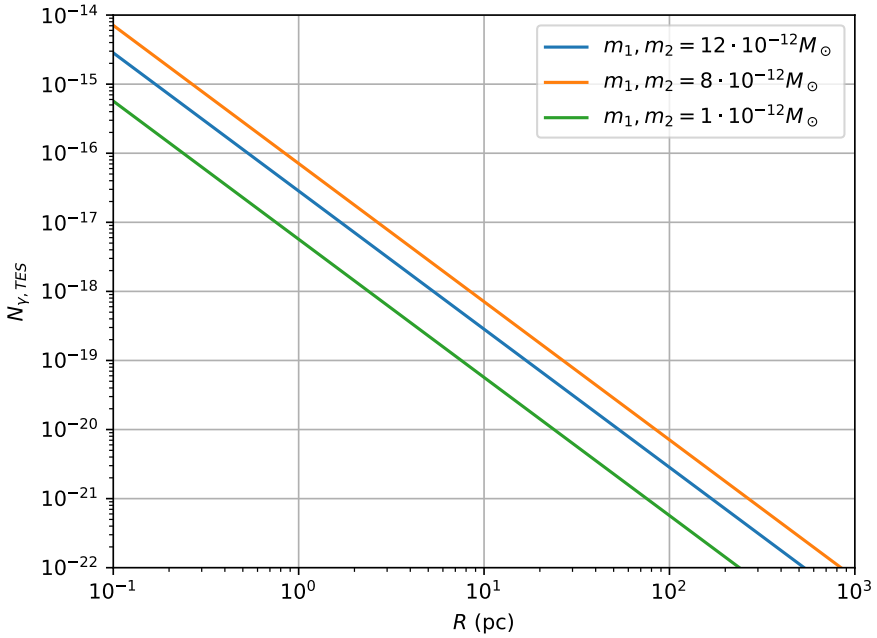


Figure 5.4: Number of detected photons $N_{\gamma,\mathrm{TES}}$ (Eq. (5.2.9)) for three equal mass PBH binary mergers as a function of distance r to the observer. For each binary the rotation axis is aligned with the line of sight to the observer.

Since the number of detected photons $N_{\gamma,\mathrm{TES}}$ is proportional to $1/r^2$ it decreases rapidly over the distance. Generally higher mass PBH merger create a stronger signal because of the dependence on the critical mass ($N_{\gamma,\mathrm{TES}} \sim M_c^{5/3}$). However for mergers close to the maximum detectable mass of $1.32 \cdot 10^{-11} M_\odot$ the limit on the frequency $f_{\max}$ for the innermost stable circular orbit can cause them to have a weaker signal than those with a smaller chirp mass.

To determine whether or not ALPS II is sensitive enough to detect these photons the detection significance S needs to be analysed. In Ref. [20] an application for background rejection for the TES using machine learning is presented. The detection significance S which is defined as

$$S = 2\sqrt{T}\left(\sqrt{\epsilon_d \epsilon_a n_s + n_b} - \sqrt{n_b}\right), \tag{5.2.10}$$

is given for different machine learning algorithms and a previous cut based analysis. Here T is the observation time, ϵ_a, ϵ_d are the analysis and detection efficiency respectively, n_s is the expected signal rate and n_b is the expected background rate. The analysis in Ref. [20] is done specifically to distinguish between background photons and 1064 nm signal photons. For the case of gravitational waves however, the entirety of the frequency spectrum that can be observed with the TES is of interest. Thus in theory a background rejection analysis would have to be made to also account for signal photons that are not 1064 nm. This would most likely prove fairly difficult.

So for this analysis the values for the background rate $n_b = 6.9 \cdot 10^{-6}$ Hz and analysis efficiency of $\epsilon_a = 0.898$ for the cut based analysis from [20] will be used. Note that this will most likely underestimate the background rate by a significant amount. The total number of signal photons $N_{\gamma,\mathrm{TES}}$ can be expressed through the signal rate n_s as

$$N_{\gamma,\mathrm{TES}} = \epsilon_d n_s T \tag{5.2.11}$$

Introducing this into Eq. (5.2.10) gives the expected detection significance for the gravitational wave signal:

$$S = 2\left(\sqrt{N_{\gamma,\mathrm{TES}} + n_b T} - \sqrt{n_b T}\right) \tag{5.2.12}$$

The detection significance S determines the sensitivity of the ALPS II experiment to the signal of PBH mergers. So for a particular PBH binary the question arises at what distance from earth the detection significance is sufficiently large so the merger can be observed. For the case of a the binary merger with masses $m_1, m_2 = 8 \cdot 10^{-11} M_\odot$ the significance as a function of the distance r to the observer is shown in Figure 5.5 for different times of observation.

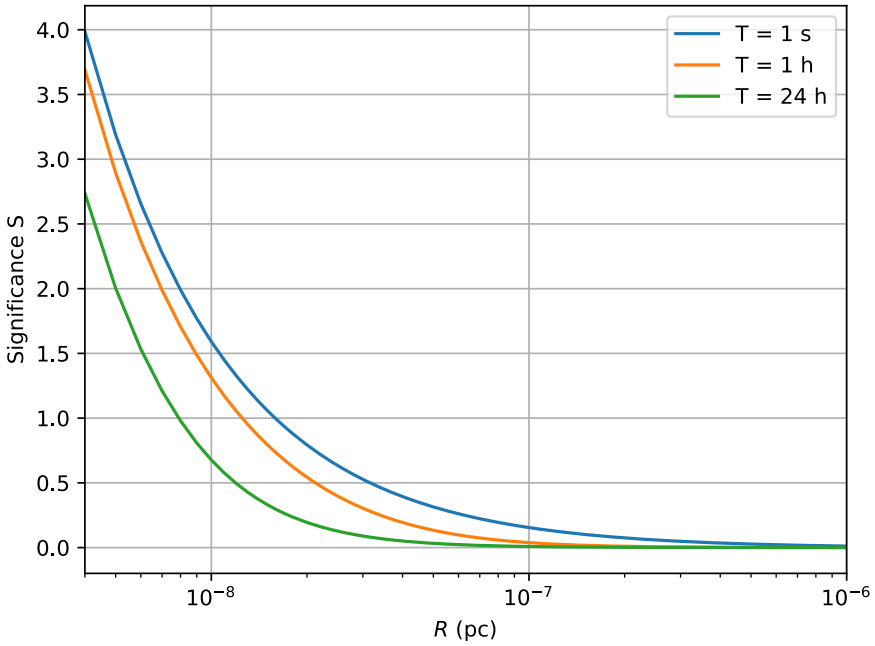


Figure 5.5: Detection significance for the gravitational wave signal of an equal mass PBH merger with $m_1, m_2 = 8 \cdot 10^{-11}$ as a function of the distance to the observer.

As can be seen even for short observation times the distance for which the significance reaches a minimum of 2 needs to be well below 10^{-8} pc. This would amount to a distance of about $3 \cdot 10^8$ m, which is even smaller than an astronomical unit (AU $\approx 1.5 \cdot 10^{11}$ m). This observation proves that the considered ALPS II experiment with the TES as a detector is not suited to detect PBH mergers. Even for the best case with the gravitational wave entering the optical cavity so it is exactly perpendicular to the magnetic field maximising the GRAPH mixing the signal is too weak to be observed for distances r outside of the solar system. Additionally the number of background

events n_b is even being underestimated, leading to an overestimation of the significance. Now it should be mentioned that in the computation of $N_{\gamma,\mathrm{TES}}$ the optical resonators in the Regeneration Cavity with the power build of $\beta_{RC} = 40000$ is not being accounted for. However, even the factor of 40000 would not change the outcome of the sensitivity outside the solar system.

6 Conclusion and Prospects

In Chapter 2 the gravitational wave strain (or more precisely its Fourier transform) of a PBH merger was calculated. Following that, the conversion of gravitational waves inside a homogenous magnetic field was computed (see Chapter 3). For this calculation it is assumed that the gravitational wave propagates along the z-axis of a Cartesian coordinate system perpendicular to the magnetic field. Both these results were combined in Chapter 5 to compute the number of photons that a gravitational wave produces when entering the constant magnetic field of the Reproduction Cavity at ALPS II. Again it is assumed that the gravitational wave enters perpendicular to the magnetic field which is the optimal case maximising GRAPH-mixing.

The number of photons $N_{\gamma,\mathrm{TES}}$ that could be absorbed by the TES were estimated by modelling the quantum efficiency of the TES after the measured reflectance from Ref. [9]. Additionally losses through the optical fiber are estimated to contribute to a further reduction of the photon signal by 0.46. Finally the number of photons is used to determine the significance S of the signal. The assumed background rate n_b and analysis efficiency of the background rejection are taken from [20]. Since the background rejection is specifically to detect signal photons of $\lambda = 1064$ nm it is assumed that for this analysis the background rate is significantly underestimated.

Even though this is the case, the analysis showed that for the significance to be $S > 2$ allowing for a detection, the PBHs would have to be inside of the solar system. Thus it is concluded that PBH mergers can not be detected by the ALPS II experiment with the TES as the detector. The expected photon signal $N_{\gamma,\mathrm{TES}}$ is so small that possible improvements in efficiency or rejection of background events, be it with the TES or possibly other detectors like the SNSPD, will most likely not change this circumstance. So for future searches of PBH other experiments might be better suited, for example searches for Hawking radiation from evaporating black holes.
On a positive note both the TES and SNSPD will probably find good application in the future of the ALPS II experiment and for axion experiments in general. The analysis and optimisation of these single photon detectors gives thus a lot of possibility for future research.

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